

NEHA GUPTA

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EDUCATION

University of Georgia	Athens, GA
<i>Doctor of Philosophy (PhD in Bioinformatics)</i>	GPA: 4.00/4.00 Aug 2023 - Present
Boston University	Boston, MA
<i>Master of Science in Bioinformatics</i>	GPA: 3.65/4.00 Sep 2020 - Jan 2022
Amity University	Noida, India
<i>Bachelor of Technology in Biotechnology</i>	GPA: 7.61/10.00 Jul 2015 - Jun 2019

TECHNICAL SKILLS

Bioinformatics Techniques: Single cell RNA-seq (scRNA), Single nucleus RNA-seq (snRNA), Spatial transcriptomics, Whole genome and Variant calling analysis, Bulk RNA-seq, pipeline development, Protein structural analysis, Drug discovery & Molecular docking,

Bioinformatics Tools: Cellranger, STAR, GATK, BWA

R & Bioconductor Packages: Seurat, DESeq2, dplyr, tidyverse, ggplot2, enrichR, stringr, pubmed.mineR, Biomart, Harmony

Python modules: Scanpy, AnnData

High-Performance Computing: Linux-based HPC environments, SLURM workload manager, Cluster Computing (AWS), job scheduling and resource management

EXPERIENCE

<i>Graduate Research Assistant</i>	University of Georgia, Athens, GA	Aug 2023 - Present
Advisor: Dr. Natarajan Kannan		

- Conducting single-cell transcriptomic and isoform-level analyses of neuronal cells to resolve cell-type-specific kinase expression and splicing in neurodegenerative disease
- Applying a single-cell co-expression framework to identify kinase–substrate networks altered in Alzheimer's disease (manuscript in preparation)
- Characterized kinase regulatory motifs and catalytic features through structural modeling with AlphaFold3 and PyMOL, contributing to a published study of the neurodegeneration-linked DCLK3 kinase (J. Biol. Chem., 2025)
- Built reproducible scRNA-seq and whole-genome variant-calling pipelines on a SLURM-managed HPC cluster for UGA's Computational Biology Consulting Group, supporting analyses for campus research labs
- Contributing to a multi-agent LLM system project that predicts cancer mutations affecting phosphorylation sites and generates testable kinase–substrate specificity hypotheses (manuscript in preparation)
- Mentored an NSF REU undergraduate through a summer research project in bioinformatics (2025)

<i>Research Associate II, Bioinformatics</i>	Cellarity Inc., Somerville, MA	Feb 2022 - Jun 2023
Manager: Dr. Scott Steelman		

- Analyzed single-cell transcriptomic and Illumina sequencing data to support laboratory operations and prototype new workflows in a drug-discovery setting
- Designed and implemented QC modules for sequencing data, catching low-quality samples and runs before downstream analysis
- Managed and maintained bioinformatics data-processing pipelines across ongoing projects, working closely with wet-lab scientists to deliver analysis-ready data

Advisor: Dr. Ayshwarya Subramanian

- Analyzed single-nucleus and single-cell RNA-seq data from TIM-3 (Havcr2)-deficient 5xFAD Alzheimer's mouse models, helping define a microglial subpopulation with increased pro-phagocytic and reduced pro-inflammatory gene expression; co-author, Kimura et al., *Nature* 2025
- Performed preprocessing, QC, integration, clustering, differential expression and functional annotation across snRNA-seq, and bulk RNA-seq datasets from in vitro and in vivo disease models
- Supported a multi-omics study of the lipid kinase PI5P4Ky through integrative analysis and visualization of transcriptomic and chromatin data

Advisor: Anuraj Nayariseri

- Implemented transcriptome expression analysis pipeline using HISAT2 and StringTie for 9 paired normal and colon cancer tissue samples, performing differential gene expression analysis with DESeq2

PUBLICATIONS

- Kimura, K., Subramanian, A., Yin, Z., Khalilnezhad, A., Wu, Y., He, D., ... **Gupta, N.**, ... & Kuchroo, V. K. (2025). Immune checkpoint TIM-3 regulates microglia and Alzheimer's disease. *Nature*, 641(8063), 718–731.
- Lu, J. D., Zhao, P., Prasad, A., **Gupta, N.**, Gravel, N., Shidhaye, T., ... & Kannan, N. (2025). Deciphering the unique autoregulatory mechanisms and substrate specificity of the understudied DCLK3 kinase linked to neurodegenerative diseases. *Journal of Biological Chemistry*, 301(10).
- **Gupta N**, Choudhary SK, Bhagat N, Karthikeyan M, Chaturvedi A. (2021). In Silico Prediction, Molecular Docking and Dynamics Studies of Steroidal Alkaloids of *Holarrhena pubescens* Wall. ex G. Don to Guanylyl Cyclase C: Implications in Designing of Novel Antidiarrheal Therapeutic Strategies. *Molecules.*; 26(14), 4147.

AWARDS

- Beverly B. Ledakowski Cancer Research Fellowship (2026)
- Student Leader of the Year, International Student Life, University of Georgia (2025)
- Merit-Based Scholarship, Boston University (2020)

PRESENTATIONS

- Poster Presentation, American Society of Human Genetics (ASHG) Annual Meeting, Boston, MA (2025)
- Poster presentations at national and international life sciences and bioinformatics conferences, India

LEADERSHIP

- President, Indian Student Association, University of Georgia (2024–2025)
- Member, Faculty Speaker Seminar Nomination Committee, Institute of Bioinformatics, University of Georgia